Calculation of $(I - A)^{-1}D^{1/2}$

Let $\tilde{L}$ be an $n \times n$ lower triangular matrix
for all $i \in 1, ..., n$ **do**
  for all $k \in i, ..., n$ **do**
    Let $l = A^T_{\mathcal{N}_i, i} \tilde{L}_{\mathcal{N}_i, k}$
    if $i = k$ **then**
      $\tilde{L}_{i,k} = (1 + l)\sqrt{D_{kk}}$
    else
      $\tilde{L}_{i,k} = l\sqrt{D_{kk}}$
    end if
  end for
end for